G4 AI MASTER CHALLENGE 001 / BUSINESS DIAGNOSIS

RavenStack - Beyond the Churn Headline

A monthly, evidence-driven diagnosis of growth, retention and commercial movement

The count rose. The customer-retention answer is more complicated.

A decision paper grounded in five source tables, a daily account-plan reconstruction and a monthly executive view.

CONDITIONAL LIFECYCLE RECONSTRUCTION

Prepared by Thomas Leick

Independent submission · 22 October 2026 · No G4 affiliation implied



OBSERVE



DECOMPOSE



PERSIST

</>

BUILD

COFFEE / CUBE / DISCIPLINE / CODE

EVIDENCE BEFORE CERTAINTY

Q1 / ROOT CAUSE

The headline is not yet a churn rate.

No causal root cause is established. The strongest verified finding is a conflict among outcome indicators and lifecycle records.

COMMERCIAL CUSTOMER CHURN

UNKNOWN

No reconciled customer outcome / eligible opening population. Conditional ledger result is shown separately.

2024 PLAN-PROXY GROSS LOSS

1.87%

7 / 374 opening account x plan-tier relations lost coverage at least once.

PLAN PROXY STILL LOST AT CLOSE

0.80%

3 / 374 opening relations remained without that plan-tier coverage at year close.

2024 OPENING CUSTOMERS LOST

0 / 190

No opening-cohort account lost all reconstructed plan coverage; this is not certified commercial churn.

THE FINDING

A customer may lose one plan-tier relationship and remain active through another. Across 2023-2024, 54 of 62 date-derived plan losses occurred while another plan remained active. Eight episodes reached zero plan-proxy coverage; all eight later returned within the observed horizon.

LEADERSHIP DECISION

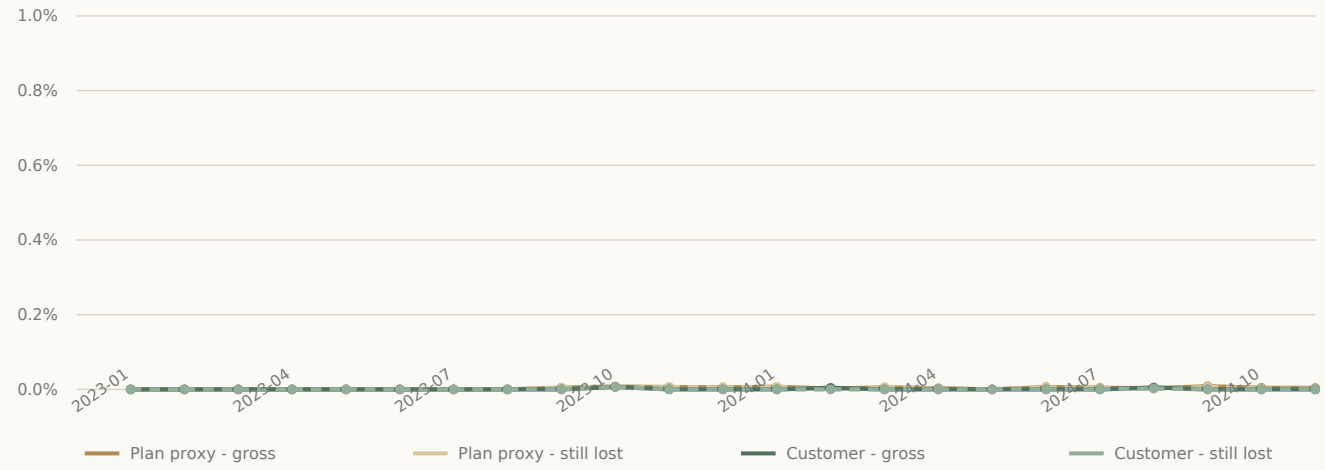
Reconcile lifecycle and event linkage before funding a broad retention campaign. Review the 60 accounts with reconstructed plan loss, starting with the 8 full-coverage-loss/return cases. Run a controlled test after defining eligible customers and a verified 90-day outcome.

02 / EXPOSURE, FLOWS AND RATE

More losses can accompany a larger base.

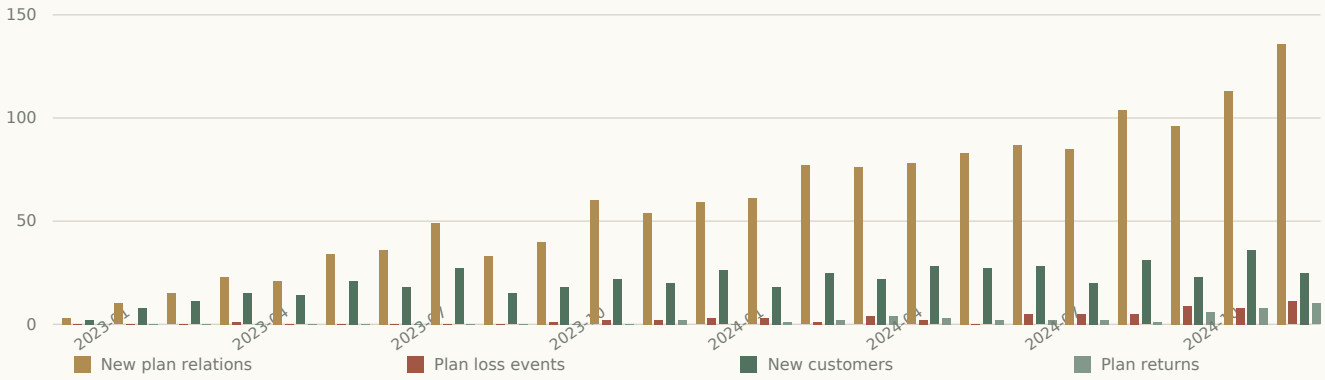
The monthly ledger is the executive series. Counts, exposure and normalized indicators answer different questions.

MONTHLY OPENING-COHORT LOSS RATES · 24 MONTHS



Solid = opening-cohort relations/customers losing coverage during month. Dashed = still lost at month close. Rates use the matching opening plan proxy or unique-account base; Jan 2023 is undefined because opening denominators are zero.

MONTHLY MOVEMENTS · COUNTS, NOT RATES



Recorded events / 100 account-months

8.64 -> 13.48

Plan losses / 100 relation-months

0.427 -> 0.595

Complete-loss episodes / 100 account-months

0.237 -> 0.153

2023 -> 2024: event rows 73 -> 527; opening exposure 845 -> 3,909 account-months (4.63x); plan losses 6 -> 56 across 1,406 -> 9,414 relation-months. A symmetric arithmetic split attributes about 339 additional event rows to exposure and 115 to event intensity. This is not causal attribution; recording change is not identifiable.

03 / CROSS-TABLE DIAGNOSIS

The records explain the disagreement, not its cause.

The five tables were compared at their natural grain; account outcomes were deduplicated before cross-table summaries.

ACCOUNTS · 500	110 account churn flags. Compared with any event history: 75 both, 35 flag-only, 277 event-only, 113 neither; 312 accounts differ pairwise.
SUBSCRIPTIONS · 5,000	5,000 unique subscription IDs; 529 upgrade-flag rows and 218 downgrade-flag rows, with 23 rows carrying both. No linked from/to plan change or effective transition record.
FEATURE USAGE · 25,000	Source usage count 124,561 -> 125,964 (+1.13%); per observed account-month 24.462 -> 24.747 (+1.17%). 19,142 rows predate linked subscription start; 290 follow an end date.
SUPPORT · 2,000 TICKETS	825/2,000 satisfaction scores missing (41.25%). Any-event vs no-event mean: 3.9804 (815 scored; 580 missing) vs 3.9833 (360; 245 missing). Escalation: 71/1,395 (5.09%) vs 24/605 (3.97%).
CHURN EVENTS · 600	73 rows in 2023; 527 in 2024; 352 accounts; 339 non-reactivation rows. Reasons: features 114, support 104, budget 104, unknown 95, competitor 92, pricing 91.

WHAT IS SUPPORTED

Usage totals rose slightly while the observed account-month denominator was nearly unchanged; that is descriptive source volume, not verified adoption by active customers. Ticket satisfaction is very similar across retrospective event labels, but missing scores and unknown event timing limit the comparison. Reason codes can prioritize customer interviews; they do not prove root cause.

RECORDED MONEY FIELDS

The subscription file includes `mrr_amount` and `arr_amount` for all 5,000 rows; 778 rows are zero and ARR equals 12 x MRR on every row. Yet the source does not provide a reconciled account billing ledger or a validated effective-period/change mapping. Do not add repeated contract-row values and call the sum lost revenue. Revenue contraction remains not identified.

Q2 / WHO NEEDS ATTENTION

A historical case list is not a risk model.

All rates below are conditional date-derived plan-tier proxy comparisons, not predictive probabilities.

2024 OPENING PLAN-RELATION COHORT

Tier proxy	Opening	Lost in year	Rate	Still lost at close
Enterprise	**120**	**5**	**4.17%**	**2**
Basic	**132**	**2**	**1.52%**	**1**
Pro	**122**	**0**	**0.00%**	**0**

Industry exposure (conditional plan-loss rows / relation-months) also varied: DevTools 10/2,215 (0.451 per 100); FinTech 8/2,146 (0.373); HealthTech 9/1,937 (0.465); Cybersecurity 9/1,716 (0.524); EdTech 6/1,400 (0.429). These are exposure diagnostics, not annual customer churn rates or causal rankings.

EIGHT RECONCILIATION CANDIDATES

Each account below had a reconstructed complete-coverage-loss episode followed by later date-derived coverage. Ask CS/RevOps to confirm whether the customer actually left, paused or changed contracts.

A-180abf
DevTools / US / event / Pro
2 plan losses; 1 full-loss episode

A-0a282f
DevTools / US / organic / Basic
1 plan loss; 1 full-loss episode

A-0baac2
FinTech / DE / organic / Enterprise
1 plan loss; 1 full-loss episode

A-40a557
Cybersecurity / AU / partner / Pro
1 plan loss; 1 full-loss episode

A-410e86
FinTech / US / organic / Basic
1 plan loss; 1 full-loss episode

A-432483
Cybersecurity / US / ads / Pro
1 plan loss; 1 full-loss episode

A-8adedc
EdTech / US / event / Basic
1 plan loss; 1 full-loss episode

A-c9bb3e
EdTech / US / ads / Pro
1 plan loss; 1 full-loss episode

These are account IDs for follow-up, not validated future-risk labels. The source event table has account_id but no subscription_id, so the event-to-plan relation remains unproven.

Q3 / WHAT SHOULD RAVENSTACK DO?

Five moves. One measurement gate.

The investment sequence follows evidence readiness: correct lifecycle definitions, reconcile cases, then test retention.

01

Instrument lifecycle

Data / Product Analytics · 0-30 days

Link cancellation, reactivation and plan change to account + subscription IDs and effective timestamps. KPI: linked, reconciled lifecycle events.

02

Review historical cases

Customer Success + RevOps · 0-30 days

Review 60 accounts with any plan loss, starting with the 8 full-coverage-loss/return cases. KPI: customer-confirmed outcome and reason documented.

03

Audit transition flags

RevOps / Product · 30-60 days

Reconcile 529 upgrade and 218 downgrade rows, including 23 with both. KPI: dated source/destination plan and comparable contract amount.

04

Measure activation

Product Analytics · 30-60 days

Instrument a canonical first-value event and meaningful adoption per eligible account cohort. KPI: 7/30-day instrumentation coverage.

05

Run a holdout test

Customer Success + Analytics · 60-90 days

Randomize eligible accounts to outreach vs. holdout after definitions are stable. Primary metric: verified customer coverage at 90 days.

IMPACT SENSITIVITY - NOT REVENUE OR CAUSAL ROI

Reference population: 60 historical accounts with a reconstructed plan loss; current eligibility is unknown. If all 60 were contacted, retained-account equivalents = 60 x assumed uptake x assumed incremental lift.

Case	Assumed uptake x lift	Retained equivalents	Gross benefit	Contact cost	Break-even monthly value
Conservative	50% x 5 pp	1.5	1.5 x V x H	60 x C	40 x C / H
Base	70% x 10 pp	4.2	4.2 x V x H	60 x C	14.29 x C / H
Optimistic	90% x 15 pp	8.1	8.1 x V x H	60 x C	7.41 x C / H

V = assumed monthly contribution per retained account; H = benefit horizon in months; C = assumed contact cost per account. These parameters are not estimated from RavenStack's records. Net modeled contribution = gross benefit - contact cost. No verified MRR, LTV or treatment effect is claimed.

06 / METHOD, LIMITS AND LABORATORY

Use the model to ask better questions.

The paper's measures are deliberately explicit about unit, period and evidence status.

CLASSIC CUSTOMER CHURN

Unique eligible customers lost in the month / unique active customers at month opening x 100. This commercial KPI is NOT IDENTIFIABLE as verified because supplied statuses, event records and date-derived coverage conflict.

PLAN-COVERAGE PROXY

Unit = account x plan_tier. Monthly gross opening-cohort loss = distinct opening relations losing coverage at least once / opening relations x 100. Persistent closing loss uses opening relations still inactive at month close. This is not product churn.

TIME AND CENSORING

Daily ledger is the source; executive reporting is monthly. Intervals use [start_date, end_date); null ends remain open to the observed horizon. End-date commercial meaning and intraday ordering are not independently validated. January 2023 opening base is zero: rate is UNDEFINED.

CAUSE AND ECONOMICS

No causal root cause or calibrated future risk was established. Refunds are not recurring revenue. Subscription-row MRR/ARR and flags lack a fully reconciled effective-dated account billing/change ledger. Revenue contraction, GRR, NRR and LTV exposure remain unknown.

DECISION EXPERIMENT

After lifecycle repair, define eligible paid accounts at risk of verified cancellation; randomize at account level to CS outreach or holdout; measure account coverage at 90 days, plan contraction and support guardrails. Report intention-to-treat difference and confidence interval; avoid mixing ticket-level records as independent customers.

THE MONTHLY BUSINESS LABORATORY

The local React + FastAPI application is a companion for observed monthly flows and transparent scenarios. Run from the standalone package: cd solution && docker compose up --build. API: http://localhost:8000 · web: http://localhost:5173. Source analyses, data provenance, tests and the chronological process log remain inside this submission.

SOURCES

G4 Challenge 001: github.com/Gestao-Quatro-Ponto-Zero/ai-master-challenge/tree/main/challenges/data-001-churn

Submission guide and template: github.com/Gestao-Quatro-Ponto-Zero/ai-master-challenge

SaaS Subscription & Churn Analytics, Kaggle, synthetic; credit River @ Rivalytics; MIT license per dataset/challenge metadata.